

Project 4: Variant Calling with Deep Learning

Goal of the Project

Students will learn how to:

- Represent sequencing data
- Understand the variant calling problem
- Use a deep learning-based approach to identify variants
- Analyze outputs from a trained variant caller

Biological Background

Variant calling is the task of identifying differences between a sample genome and a reference genome, such as: SNPs (single nucleotide polymorphisms) and Insertions/deletions (indels). Input data typically comes from Aligned sequencing reads (BAM/CRAM files).

Lab 1 — Introduction to Variant Calling

- Objective
 - Understand what a genetic variant is
 - Explore reference vs sample differences
- Guided Questions
 - A. What is a SNP?
 - B. What is the difference between a reference genome and a sample genome?
 - C. Why is variant calling important in genomics?

Lab 2 — Understanding Sequencing Reads

- Objective
 - Simulate or inspect sequencing reads
 - Understand alignment conceptually
- Code: *understanding-reads.py*
- Guided Questions
 - A. What is a sequencing read?
 - B. Why do reads need to be aligned to a reference genome?
 - C. What kinds of errors or variations might appear in reads?

Lab 3 — Representing Reads as Features

- Objective
 - Convert nucleotide sequences into numerical representations
- Code: *representing-reads.py*
- Guided Questions
 - A. Why use one-hot encoding for DNA sequences?
 - B. How does this representation compare to word embeddings in NLP?
 - C. What are limitations of one-hot encoding?

Lab 4 — Understanding DeepVariant Conceptually

- Objective
 - Understand how deep learning is used in variant calling
- Key Idea → deepVariant works by:
 - Converting aligned reads into image-like or tensor-like representations
 - Feeding them into a convolutional neural network
 - Predicting: genotype likelihoods, variant presence
- Guided Questions
 - A. Why is variant calling framed as a classification problem?
 - B. What is the input and output of a deep learning variant caller?
 - C. Why might deep learning outperform rule-based methods?

Lab 5 — Evaluating Variant Calls

- Objective
 - Compare predicted variants with ground truth
- Code: *evaluating-variant-calls.py*
- Guided Questions
 - A. What does precision measure?
 - B. What does recall measure?
 - C. Why is evaluation important in variant calling?
 - D. What challenges arise in defining “ground truth” in genomics?

Lab 6 — End-to-End Pipeline Concept

- Objective
 - Connect all steps into a full workflow
- Pipeline Overview
 - Raw sequencing reads
 - Alignment to reference genome
 - Tensor/image representation of reads
 - Deep learning model (DeepVariant)
 - Variant predictions
 - Post-processing and evaluation
- Guided Questions
 - A. What are the key stages of a variant calling pipeline?

- B. Where does deep learning fit into the pipeline?
- C. What are the advantages of automation in this workflow?
- D. How does this differ from traditional bioinformatics pipelines?